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Flexible chain implants and instrumentation

Abstract

A flexible chain implant for insertion into an interior volume of a vertebral body. The implant may be implanted in an insertion position for sliding through a cannula and is flexible for packing into the interior volume in an implanted configuration. The implant randomly separates in the implanted configuration. The implant includes a top member and a bottom member, wherein the top and bottom members are coupled to one another at a coupled portion. The top and bottom members preferably each include an inner surface such that the inner surfaces include a plurality of alternating projections and recesses so that the projections are received within the recesses in an insertion position. Alternatively, the implant may include a plurality of substantially non-flexible bodies and a plurality of substantially flexible links interconnecting the bodies. The non-flexible bodies include a plurality of facets and/or abutment surfaces.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of pending U.S. patent application Ser. No. 15/869,442, filed Jan. 12, 2018, which is a continuation of U.S. patent application Ser. No. 14/165,979, filed Jan. 28, 2014, now U.S. Pat. No. 9,907,667, issued Mar. 6, 2018, which is a divisional of U.S. patent application Ser. No. 12/601,468, filed Nov. 23, 2009, now U.S. Pat. No. 8,673,010, issued Mar. 18, 2014, which is a national phase under 35 U.S.C. § 371 of PCT Application No. PCT/US2008/068836, filed Jun. 30, 2008, which claims the benefit of U.S. Provisional Application No. 60/947,258, filed on Jun. 29, 2007. The entire disclosure of each application listed in this paragraph is incorporated by reference in this application for all purposes.

TECHNICAL FIELD

(1) The invention relates to implants, and more particularly to implants for augmenting or supporting bones or other structures, such as, for example, a vertebral body.

BACKGROUND

(2) Vertebral compression fractures (“VCF”) represent a common spinal injury and may result in prolonged disability. Generally speaking, VCF involves collapsing of one or more vertebral bodies in the spine. VCF usually occurs in the lower vertebrae of the thoracic spine or the upper vertebrae of the lumbar spine. VCF generally involves fracture of the anterior portion of the affected vertebral body. VCF may result in deformation of the normal alignment or curvature, e.g., lordosis, of the vertebral bodies in the affected area of the spine. VCF and/or related spinal deformities may result, for example, from metastatic diseases of the spine, from trauma or may be associated with osteoporosis. Until recently, doctors were limited in how they could treat VCF and related deformities.

(3) Recently, minimally invasive surgical procedures for treating VCF have been developed. These procedures generally involve the use of a cannula or other access tool inserted into the posterior of the targeted vertebral body, usually through the pedicles.

(4) In one such procedure, a cannula or bone needle is passed through the soft tissue of the patient's back. Once properly positioned, a small amount of polymethylmethacrylate (PMMA) or other orthopedic cement is pushed through the needle into the targeted vertebral body. This technique may be effective in the reduction or elimination of fracture pain, prevention of further collapse, and a return to mobility in patients. However, this technique typically does not reposition the fractured bone into its original size and/or shape and, therefore, may not address the problem of spinal deformity due to the fracture.

(5) Other treatments for VCF generally involve two phases: (1) reposition, or restoration of the original height of the vertebral body and consequent lordotic correction of the spinal curvature; and (2) augmentation, or addition of material to support or strengthen the fractured or collapsed vertebral body.

(6) One such treatment involves inserting, through a cannula, a catheter having an expandable

member into an interior volume of a fractured vertebral body, wherein the interior volume has a relatively soft cancellous bone surrounded by fractured cortical bone therein. The expandable member is expanded within the interior volume in an attempt to restore the vertebral body towards its original height. The expandable member is removed from the interior volume, leaving a void within the vertebral body. PMMA or other filler material is injected through the cannula into the void to stabilize the vertebral body. The cannula is then removed and the cement cures to augment, fill or fix the vertebral body.

(7) Another approach for treating VCF involves inserting an expandable mesh graft balloon, or containment device, into the targeted vertebral body. The graft balloon remains inside the vertebral body after it is inflated with PMMA or an allograft product, which limits intraoperative loss of height of the repositioned endplates.

(8) In some cases of fractured or otherwise damaged bones, bone grafts may be used to repair or otherwise treat the damaged area. In the United States alone, approximately half a million bone grafting procedures are performed annually, directed to a diverse array of medical interventions for complications such as fractures involving bone loss, injuries or other conditions necessitating immobilization by fusion (such as for the spine or joints), and other bone defects that may be present due to trauma, infection, or disease. Bone grafting involves the surgical transplantation of pieces of bone within the body, and generally is effectuated through the use of graft material acquired from a human source. Human graft material is primarily utilized due to the limited applicability of xenografts, e.g., transplants from another species.

(9) Many orthopedic procedures involve the use of allografts, which are bone grafts from other human sources (normally cadavers). Allografts, for example, are placed in a host bone and serve as the substructure for supporting new bone tissue growth from the host bone.

(10) The various bones of the human body such as the femur (thigh), tibia and fibula (leg), humerus (upper arm), radius and ulna (lower arm) have geometries that vary considerably. The lengths of these bones are varied, as well as the shape of the cross section of each type of bone and the shape of any given bone over its length. In addition, the wall thickness may vary in different areas of the cross-section of each bone. Thus, the use of any given bone to produce an implant or a component of an implant may be a function of the donor bone's dimensions and geometry. Machining of bones, however, may permit the production of an implant or a component of an implant with standardized or custom dimensions. Further, the availability of allograft bone source material is limited and the ability to enhance the bone yield from the available supply of allograft bone material is desirable.

(11) As shown in FIG. 9 and as generally described in U.S. patent application Ser. No. 11/633,131, entitled "Flexible Elongated Chain Implant and Method of Supporting Body and Tissue With Same", the entire contents of which are hereby incorporated by reference, a prior art flexible chain implant **1000** may include a plurality of bodies **1010** and a plurality of linking portions **1020** (sometimes referred to as struts, bridges or links). The implant **1000** comprises a single flexible monolithic chain having a first end **1002** and a second end **1004** formed of allograft cortical bone having a plurality of substantially non-flexible bodies connected by substantially flexible links. Such an implant **1000** may be utilized to treat VCF.

(12) Thus, it is desirable in the art to provide safe and effective implants and methods for aiding and/or augmenting fractured or otherwise damaged vertebral bodies and other bones, preferably implants that may be inserted via a minimally invasive surgical technique. Moreover, where the implant is formed from bone, it is desirable to provide implants that are designed to enhance existing bone yield while minimizing discarded excess material.

SUMMARY

(13) The present invention is preferably directed to an implant for insertion into an interior volume in a targeted vertebral body. The implant is preferably inserted into the vertebral body in a relatively compact insertion position and is oriented in an implanted configuration when located in the vertebral body so that the implant augments at least a portion of the interior volume of the

targeted vertebral body so that large pockets or voids are limited within the interior volume, while small voids amenable to bone in-growth and providing a vascular pathway are provided. The implant preferably separates or unwraps from the insertion position and, generally, randomly coils and wraps or twists around itself and packs into the implanted configuration. The implant is preferably sized and configured to be inserted into the targeted vertebral body via a minimally invasive surgical technique, such as, for example, through one or more cannulas via a transpedicular or extrapedicular approach.

(14) In one exemplary embodiment, the implant has a coupled portion and a terminal end. The implant preferably also includes a top member and a bottom member wherein the top and bottom members are coupled to one another at the coupled portion. The top and bottom members include an inner surface. Each of the inner surfaces of the top and bottom members include a plurality of alternating projections and recesses so that, when in an insertion position, the projections formed in the top member are received within the recesses formed in the bottom member and the projections formed in the bottom member are received within the recesses formed in the top member. In the insertion position, the implant is preferably sized for receipt within a cannula. The top and bottom members preferably separate from one another to an implanted configuration as the implant is being inserted into and/or when the implant is positioned within the interior volume of the targeted vertebral body so that large pockets or voids are limited within the interior volume. Preferably the implant is constructed of bone, but may be constructed of a synthetic material.

(15) In another exemplary embodiment, the implant includes a plurality of generally non-flexible bodies or nodules and a plurality of substantially flexible links interconnecting the plurality of bodies. Preferably, the bodies and links are connected end-to-end to form the implant. Each of the generally non-flexible bodies preferably includes a plurality of facets. The plurality of non-flexible bodies define a body axis and the plurality of links defining a link axis, each in an extended position, the body axis is off-set from the link axis in the extended position. Preferably the implant is formed from bone, but may be constructed of a synthetic material.

(16) In another exemplary embodiment, the implant includes a plurality of generally non-flexible bodies or nodules and a plurality of substantially flexible links interconnecting the plurality of bodies. The plurality of links is preferably connected at their ends to one of the plurality of bodies to form the implant. Each of the plurality of bodies preferably includes a tapered surface having a wide end, a narrow end, and an abutment surface. The plurality of bodies define a longitudinal body axis and the plurality of links define a longitudinal link axis in an extended position. The longitudinal link axis is off-set from the longitudinal body axis. The abutment surface formed on each one of the plurality of bodies preferably further defines a longitudinal abutment axis. The longitudinal abutment axis is off-set from the longitudinal body axis and from the longitudinal link axis. Preferably the implant is formed from bone, but may be constructed of a synthetic material.

(17) Other objects and features of the present invention will become apparent from the following detailed description, considered in conjunction with the accompanying drawing figures. It is to be understood, however, that the drawings are designed solely for the purpose of illustration and not as a definition of the limits of the invention, for which reference should be made to the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description of the preferred embodiments of the application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustrating the flexible chain implant of the present application, there is shown in the drawings preferred embodiments. The drawings are prepared

from three-dimensional models of preferred embodiments of the flexible chain implant and related tooling and are accurate for making at least dimensional comparisons between various features of the preferred implants and related components. It should be understood, however, that the application is not limited to the precise arrangements and instrumentalities shown. In the drawings:

- (2) FIG. 1 is a top perspective view of a first preferred embodiment of a flexible chain implant having external cuts and a groove for receiving a rod;
- (3) FIG. 2A is a side perspective view of a second preferred embodiment of a flexible chain implant having alternating projections and recesses in an insertion position;
- (4) FIG. 2B is a side elevational view of the implant shown in FIG. 2A, wherein a top member is spaced from a bottom member;
- (5) FIG. 3A is a side elevational view of a portion of bone cut to form alternating projections and recesses;
- (6) FIG. 3B is an exploded, side elevational view of the portion of bone shown in FIG. 3A;
- (7) FIG. 4A is a side perspective view of a third preferred embodiment of a multi-piece flexible chain implant, shown in an insertion position;
- (8) FIG. 4B is an exploded, side perspective view of the implant shown in FIG. 4A;
- (9) FIG. 5A is a side perspective view of a fourth preferred embodiment of a multi-piece flexible chain implant, shown in an insertion position;
- (10) FIG. 5B is an enlarged, partially exploded, side perspective view of the implant shown in FIG. 5A, showing a cap spaced from top and bottom members;
- (11) FIG. 5C is an exploded, side perspective view of the implant shown in FIG. 5A;
- (12) FIG. 6A is a side perspective view of a fifth preferred embodiment of a multi-piece flexible chain implant, shown in an insertion position;
- (13) FIG. 6B is an exploded, side perspective view of the implant shown in FIG. 6A;
- (14) FIG. 7A is a side perspective view of a sixth preferred embodiment of a multi-piece flexible chain implant, shown in an insertion position;
- (15) FIG. 7B is an exploded, side perspective view of the implant shown in FIG. 7A;
- (16) FIG. 8A is a side perspective view of a seventh preferred embodiment of a multi-piece flexible chain implant, shown in an insertion position;
- (17) FIG. 8B is an exploded, side perspective view of the implant shown in FIG. 8A;
- (18) FIG. 9 is a side perspective view of a prior art implant;
- (19) FIG. 10A is a side elevational view of an eighth preferred embodiment of a flexible chain implant, shown in an extended position;
- (20) FIG. 10B is a side elevational view of a ninth preferred embodiment of a flexible chain implant, shown in an extended position;
- (21) FIG. 11A is a side perspective view of a tenth preferred embodiment of a flexible chain implant, shown in an extended position;
- (22) FIG. 11B is an alternate, side perspective view of the flexible chain implant shown in FIG. 11A, also shown in an extended position;
- (23) FIG. 11C is another side perspective view of the flexible chain implant shown in FIG. 11A bending in a plane;
- (24) FIGS. 12A-12F depict various views of a cannula having an advancing mechanism for inserting a flexible chain implant into the interior volume of a vertebral body;
- (25) FIG. 13A is a top plan view of at least one of the flexible chain implants of the second, third, fourth, fifth or sixth preferred embodiments in an implanted configuration within a vertebral body of a vertebra; and
- (26) FIG. 13B is a side elevational view of at least one of the flexible chain implants of the second, third, fourth, fifth or sixth preferred embodiments in an implanted configuration within a vertebral body of a vertebra.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(27) Certain terminology is used in the following description for convenience only and is not limiting. The words “right”, “left”, “lower”, “upper”, “top” and “bottom” designate directions in the drawings to which reference is made. The words “inwardly” and “outwardly” refer to directions toward and away from, respectively, the geometric center of the device and designated parts thereof. The words, “anterior”, “posterior”, “superior”, “inferior” and related words and/or phrases designate preferred positions and orientations in the human body or with respect to the implant of the present application to which reference is made and are not meant to be limiting. The terminology includes the above-listed words, derivatives thereof and words of similar import.

(28) Certain exemplary embodiments will now be described with reference to the drawings. In general, such embodiments relate to a vertebral augmentation system for aiding and/or augmenting a patient's spine. As generally understood by one of ordinary skill in the art, it should be understood that while the preferred vertebral augmentation system will be described as and may generally be used in the spine (for example, in the lumbar, thoracic or cervical regions), those skilled in the art will appreciate that the vertebral augmentation system may be used for aiding and/or augmentation other parts of the body such as, for example, joints, long bones or bones in the hand, face, feet, extremities, cranium, or in nearly any bone in the human body.

(29) Referring to FIGS. 1-13B, a preferred vertebral augmentation system or flexible chain implant of the present application is directed to an implant and to a system and method for inserting the implant into a targeted vertebral body **602** of a vertebra **600**, such as, for example, one which has been subjected to a VCF. Generally, the implant is sized and configured to be inserted into an interior volume **604** of the targeted vertebral body **602**. Preferably, once inserted into the interior volume **604** of the targeted vertebral body **602**, the implant will separate, unwrap, or unravel from an insertion position to an implanted configuration wherein the implant randomly coils, twists, stacks and/or wraps onto itself so that the implant may generally augment the interior volume **604** of the targeted vertebral body **602**. The implant preferably separates and/or coils in the implanted configuration. Preferably, the implant is inserted into the targeted vertebral body via a minimally invasive surgical technique, such as, for example, through one or more cannulas **606**, as will be described in greater detail below.

(30) A first preferred embodiment of the flexible chain implant **10** is generally shown in FIG. 1. The implant **10** may be formed from a monolithic strip of material **12**. That is, the implant **10** may be formed from a single, integral strip of material **12**. The strip of material **12**, and hence the implant **10**, is preferably formed of bone, e.g., cortical bone, cancellous bone or both, but more preferably cortical bone. Thus, the implant **10** may be formed by cutting and/or machining a single, integral strip of material **12** from a piece of stock bone.

(31) The strip of material **12** preferably includes a flexible section **14**, which may be located generally in the middle of the strip of material **12**, so that the strip of material **12** may be folded over onto itself. In this manner, as shown, the implant **10** includes a top member **20** and a bottom member **30** integrally connected at a coupled portion **50**. The implant **10** may also include a gap **52** proximate the coupled portion **50**. The gap **52** facilitates the separation of the top member **20** from the bottom member **30**, for example once the implant **10** has been inserted into the interior volume **604** of the targeted vertebral body **602**, as will be described in greater detail below. The gap **52** may also facilitate bending and/or folding of the top and bottom members **20**, **30** relative to each other. When harvested or initially machined from stock bone, the flexible section **14** is generally not flexible, but is further processed, typically through demineralization, to make the flexible section **14** generally flexible, as will be described in greater detail below.

(32) Referring to FIGS. 1, 13A and 13B, in use, the implant **10** is preferably inserted into the interior volume **604** of the targeted vertebral body **602** through a minimally invasive surgical technique, such as, for example, via one or more cannulas **606** while the implant is positioned in the insertion position. Once the implant **10** has been inserted or as the implant **10** moves out of the cannula **606** into the interior volume **604** of the targeted vertebral body **602**, the top and bottom

members **20, 30** preferably separate or unwrap from their insertion position and, generally, randomly coil and wrap or twist around themselves in the implanted configuration to substantially augment the interior volume **604**. Preferably, in use, the non-integral, uncoupled or terminal end **54** of the implant **10** is inserted first into the cannula **606** and enters the interior volume **604** prior to the remainder of the implant **10**. Although as will generally be appreciated by one of ordinary skill in the art, the coupled portion **50** of the implant **10** may be inserted first into the cannula **606**. For example, the implant **10** of the first preferred embodiment may be pushed into the cannula **606** with the coupled portion **50** entering first, using a rod **80**, as will be described in greater detail below.

(33) The implant **10** may also be pulled into the interior volume **604** of the targeted vertebral body **602** through a first access, such as through a hole in a first or second pedicle **608a, 608b** via, for example, a second access hole or channel formed in the opposing one of the first or second pedicle **608a, 608b**. To facilitate pulling or pushing of the implant **10** into the interior volume **604** of the targeted vertebral body **602**, the implant **10** may include a longitudinal channel or groove **70** (as best shown in FIG. **1**) formed in the outer surface of the implant **10**. The channel **70** preferably extends along the length of the implant **10** from the coupled portion **50** to the terminal end **54** in the insertion position. As shown, the channel **70** may be formed in a side of the implant **10**. However, as will be generally appreciated by one of ordinary skill in the art, the channel **70** may be formed in any surface of the implant **10** including, but not limited to, the top or bottom surface. Alternatively, the implant **10** may include a longitudinal access hole or lumen (not shown) substantially parallel to the longitudinal axis of the implant **10** for receiving a rod for pulling the implant **10** into the interior volume **604** of the targeted vertebral body **602**.

(34) Referring to FIGS. **4A, 5A, 6A, 7A** and **8A**, the flexible chain implants **100, 200, 300, 400, 500** third, fourth, fifth, sixth and seventh preferred embodiments include a sloped or tapered surface at the terminal end **54** in the insertion position that slopes outwardly from a separation line between the top and bottom members **20, 30**, generally away from the coupled portion **50**. The sloped surface promotes separation of the top and bottom members **20, 30** from the insertion position to the implanted configuration as the implant **100, 200, 300, 400, 500** is inserted into the interior volume **604**. In addition, the nodules **41** include tapered or sloping surfaces extending toward the hinges **43** that promote the unwinding of the top and bottom members **20, 30** relative to each other as the implant **100, 200, 300, 400, 500** is urged into the interior volume **604**. The flexible chain implant **100, 200, 300, 400, 500** is not limited to inclusion of these tapered and/or sloping features and may function without these features. In addition, the tapered and sloping features may be included in any of the preferred embodiments of the present invention, despite whether the features are shown in the drawings of any of the particular embodiments.

(35) The channel **70** is preferably sized and configured to receive a rod **80**, which may be flexible or generally stiff. In accordance with the first preferred embodiment, the rod **80** includes an elongated portion **82** having a generally cylindrical shape and a bent portion **84**. The bent portion **84** is preferably located adjacent an end of the elongated portion **82**. The elongated portion **82** of the rod **80** is preferably sized and configured to be received, at least partially, within the channel **70** formed in the implant **10**, while the bent portion **84** is preferably sized and configured to be received within the gap **52** formed in the implant **10**. In this manner, both the implant **10** and the rod **80** may be received within the cannula **606**. The rod **70** preferably is coupled to the implant **10** adjacent to the coupled portion **50** so that the user may pull or push the implant **10** through the cannula **606** and into the interior volume **604** of the targeted vertebral body **602**. That is, the rod **70** preferably assists the user to pull or push the implant **10** through the cannula **606** and into the targeted vertebral body **602**. After the implant **10** is inserted into the patient's body, the rod **70** may be removed. Specifically, once the implant **10** is positioned completely within the interior volume **604** and is out of the cannula **606**, removal of the rod **80** from the interior volume **604** may promote the separating, unwrapping, uncoiling or unfolding of the implant **10** from the insertion position (FIG. **1**) to the implanted configuration (e.g. FIGS. **13A** and **13B**). The rod **80** is preferably,

generally rigid when configured to push the implant **10** into the interior volume **604** and is preferably, generally flexible or elastic when configured to pull the implant **10** into the interior volume **604**, as will be understood by one having ordinary skill in the art.

(36) In the first preferred embodiment, the top and bottom members **20**, **30** of the implant **10** includes a plurality of cuts **25** formed in the outer surface thereof in order to increase the overall flexibility of the implant **10**. Specifically, when the implant **10** of the first preferred embodiment is constructed of bone, the cuts **25** provide narrowed cross-sections along the length of the top and bottom members **20**, **30**, which may be subsequently demineralized to enhance flexibility, as will be described in greater detail below. That is, the cuts **25** define a plurality of alternating nodules **41**, which are generally comprised of the relatively thick areas between the cuts **25** of the top and bottom members **20**, **30**, and hinges **43**, which are generally comprised of portions of material at the cuts **25** of the top and bottom members **20**, **30**. The hinges **43** are preferably demineralized to the point at which they become flexible, at which point there is no or limited minerals left. However, the hinges **43** are not limited to being demineralized through their entire thickness and may be demineralized to nearly any depth that provides flexibility to the hinges **43** when the implant **10** is constructed of bone. In one exemplary embodiment, the implant **10** was constructed to have an overall length $L_{sub.1}$ of about 108 mm (about 54 mm when folded) and was sized to fit within a cannula having an inner diameter of about 4.3 mm. The nodules **41** had a length $l_{sub.1}$ of about 3.9 mm, a width $w_{sub.1}$ of about 3.4 mm, and a height $h_{sub.1}$ of about 2.0 mm. In comparison, the hinges **43** had a length $l_{sub.2}$ of about 0.6 mm, a width $w_{sub.2}$ of about 3.4 mm, and a height $h_{sub.2}$ of about 1.0 mm. In the insertion position, the flexible chain implant **10** of the first preferred embodiment has a diameter $D1$ of approximately 4.0 mm, resulting in a clearance of approximately 0.3 mm when the implant **10** is located in the 4.3 mm preferred cannula **606**.

However, as will be appreciated by one of ordinary skill in the art, the size of the cannula **606**, implant **10**, nodules **41** and hinges **43** can be varied to suit the needs of the individual patients. Accordingly, the implants of the present invention are not limited to any particular dimensions.

(37) Referring to FIGS. 2A-3B, 13A and 13B, a second preferred embodiment of a flexible chain implant **10'** includes a plurality of alternating projections and recesses **40**, **45** formed along an inner surface of the implant **10'** so that when the strip of material **12** is folded over onto itself, the projections **40** formed in the top member **20** are received within the recesses **45** formed in the bottom member **30** and the projections **40** formed in the bottom member **30** are received within the recesses **45** formed in the top member **20** in the insertion position. In this manner, the top and bottom members **20**, **30** may be nested together to minimize the size of the implant **10** in the insertion position (FIG. 2A) such that the implant **10'** is slidably receivable within the cannula **606**. The implant **10'** may then be pulled or pushed through the cannula **606** and into the interior volume **604** of the targeted vertebral body **602** via, for example, a push rod (not shown) or the rod **80**.

(38) The implant **10'** of the second preferred embodiment enables, depending on the stock material **13**, may enable multiple strips of material **12a**, **12b** to be simultaneously machined from a single, monolithic piece of stock bone **13**. That is, as generally depicted in FIGS. 3A and 3B, since nesting of the top and bottom members **20**, **30** preferably includes alternating projections and recesses **40**, **45**, two strip of materials **12a**, **12b** may be simultaneously machined from a single piece of stock bone **13**. Moreover, as will be described in greater detail below, during insertion of the implant **10'** into the interior volume **604** of the targeted vertebral body **602**, the implant **10'** may be subjected to relatively large compressive stresses as the implant **10'** is forced into the interior volume **604**. As a result of these relatively large compressive stresses, the implant **10'** may become damaged.

However, as a result of nesting, the top and bottom members **20**, **30**, in effect, form a solid implant body that is capable of more readily withstanding the compressive forces. Thus, nesting generally minimizes the possibility that the implant **10'** will be damaged during insertion.

(39) Although FIGS. 3A and 3B depict completely separating the two strip of materials **12a**, **12b** from the stock bone **13**, the implants **10**, **10'** of the first and second preferred embodiments may be

cut or machined from the bone **13**, generally in the inserted position (FIGS. **1** and **2A**) without completely separating the two strip of material **12a**, **12b** during the original harvesting or machining process.

(40) Alternatively, rather than being manufactured from a single, integral or monolithic strip of material **12**, the top and bottom members **20**, **30** may be formed from two or more strips of materials **12a**, **12b**, which are then coupled together, for example, at the coupled portion **50**. The strips of materials **12a**, **12b** may be machined or harvested from a single bone **13**, from multiple bones or from nearly any other material that is able to take on the general shape and size of the strip of material **12a**, **12b** and withstand the normal operating conditions of the implant **10**, **10'**.

Constructing a multiple-piece flexible chain implant (FIGS. **4A-8B**) may be particularly useful in situations where one is not able to obtain a single strip of material **12** of sufficient size to produce an integrally formed implant **10**, **10'**. The top and bottom members **20**, **30** may be coupled by any means now or hereafter known.

(41) Referring to FIGS. **4A** and **4B**, in a third preferred embodiment, a flexible chain implant **100** includes the top and bottom members **20**, **30** formed with the alternating projections **40** and recesses **45**, a dovetail-shaped projection **42** and a dovetail-shaped recess **47** at the coupled portion **50**. As shown, the top and bottom members **20**, **30** are coupled together by interconnecting the dovetail-shaped projection **42** and the dovetail-shaped recess **47** such that when the top member **20** and the bottom member **30** are separated or unwrapped from each other, such as, in direction **A**, the dovetail shaped projection **42** and recess **45** remain interconnected and secure the top member **20** to the bottom member **30**. The implant **100** may also include one or more pins (not shown) extending proximate or through the dovetail-shaped projection **42** and recess **47** for further securing the top and bottom members **20**, **30**. As shown, the implant **100** of the third preferred embodiment includes a single pair of dovetail-shaped interconnecting projections **42** and recesses **47**. More preferably, the dovetail-shaped projection **42** and recess **47** are formed adjacent the coupled portion **50** of the implant **100** so that, similar to the integral implant **10'** of the second preferred embodiment, the projections **40** formed in the top member **20** may be received within the recesses **45** formed in the bottom member **30** and the projections **40** formed in the bottom member **30** may be received within the recesses **45** formed in the top member **20** so that the implant **100** of the third preferred embodiment may be nested together in the insertion position. Once inserted, however, the top and bottom members **20**, **30** may separate or unwrap from their insertion position and, generally, randomly coil and wrap or twist around itself to augment the interior volume **604** of the targeted vertebral body **602** in the implanted configuration. The dovetail-shaped projection **42** and recess **47** preferably, fixedly secure the top member **20** to the bottom member **30** at the coupled portion **50** of the implant **100** even after the implant **100** has been inserted into the interior volume **604**. In this manner, the dovetail-shaped projection **42** and recess **47** operate in a manner substantially similar to the integral coupled portion **50** formed from the monolithic portion of bone or other material of the implant **10'** of the second preferred embodiment. However, the coupled portion **50** is not limited to fixedly securing the top and bottom members **20**, **30** together in the implanted configuration and may separate once the implant **100** is inserted into the interior volume **604**.

(42) Alternatively and/or in addition, the top and bottom members **20**, **30** may be coupled together by demineralizing the implant **100** of the third preferred embodiment. That is, by demineralizing the top and bottom members **20**, **30**, the top and bottom members **20**, **30** may become fixed relative to each other at the coupled portion **50** of the implant **100** of the third preferred embodiment. Thus, demineralizing the top and bottom members **20**, **30**, welds or otherwise secures the members **20**, **30** together when the members **20**, **30** are demineralized with the dovetail-shaped projection **42** and recess **47** engaged with each other. Demineralizing the implant **100** may be particularly useful in combination with a corresponding dovetail joint as traditionally dovetailed, multi-piece allograft implants have incorporated a pin to further secure the pieces together. Demineralizing the implant **100** may eliminate the requirement for any additional pin, thereby reducing part count and

generally simplifying the construction of the implant **100**. In such a demineralization process, the inner surfaces of the top and bottom members **20, 30** are masked or otherwise separated to space the inner surfaces from each other to provide exposure to the demineralizing medium and to generally prevent these inner surfaces from welding or becoming fixed to each other during demineralization. The coupled portion **50** of the implant **100** is not limited to being constructed of the described demineralized dovetail-type joint and may be constructed of nearly any two allograft surfaces positioned in facing engagement or close proximity to each other and demineralized and/or pinning, press fitting, capping or other known fastening mechanisms and methods that would be apparent to one having ordinary skill in the art.

(43) In order to demineralize the top and bottom members **20, 30** such that they are welded or fixed together, the implant **100**, etc. may be demineralized in a bath of HCl concentration for several hours to provide a desired penetration percentage into the thickness of the top and bottom members **20, 30** thus resulting in a relatively deep demineralized layer. The duration of demineralization is about fifteen minutes to about eight hours, and preferably between about 1.5 hours to about 2 hours. The HCl concentration is between about 0.5 HCl to about 2.5 HCl, and preferably between about 0.6 HCl and about 1.0 HCl. Although as will be generally appreciated by one of ordinary skill in the art, the duration of demineralization and the HCl concentration may be adjusted in order to obtain the desired levels of flexibility, to retain some rigidity in the nodules, and to obtain welding of the multi-piece implants. The implants of the present invention are not limited to any particular duration of demineralization and/or HCl concentration. When incorporating alternating projections **40** and recesses **45**, the plurality of nodules **41** generally are comprised of the relatively thick projections **40** while the hinges **43** generally are comprised of the portions of material at the recesses **45**. The penetration percent of demineralization, for nodules **41** of the top and bottom members **20, 30** is about one-half millimeter (0.5 mm) of demineralization on all sides. Hinges **43** are preferably demineralized to the point at which they become flexible, at which point there is no or limited minerals left. However, the hinges **43** are not limited to being demineralized through their entire thickness and may be demineralized to nearly any depth that provides flexibility to the hinges **43** when the implant **10', 100**, etc. are constructed of bone. It should be noted that the method of forming an implant from coupled bone using demineralization to fixedly secure the various components or members of the implant is not limited to the implants described herein and may be used to couple any multi-piece implant formed from bone now or hereafter known in the art.

(44) In exemplary second, third, fourth, fifth, sixth and seventh preferred embodiments, the top and bottom members **20, 30** of the implant **10', 100**, etc. were constructed to have an overall length L.sub.3 of about 40 mm to about 52 mm and was sized to fit within a cannula **606** having an inner diameter of about 4.3 mm. The nodules **41** had a length l.sub.3 of about 3.9 mm to about 6.7 mm, a width w.sub.3 of about 3.6 mm to about 3.8 mm, and a height h.sub.3 of about 2.5 mm to about 3.1 mm. In the insertion position, the implants **10', 100**, etc. of the second, third, fourth, fifth, sixth and seventh preferred embodiments have a diameter D.sub.3 of approximately 3.5 mm to approximately 4.0 mm. In addition, the hinges **43** preferably have a length l.sub.4 of about 3.1 mm to about 4.1 mm, a width w.sub.4 of about 3.3 mm to about 3.6 mm, and a height h.sub.4 of about 0.5 mm to about 1.0 mm. Although as will be generally appreciated by one of ordinary skill in the art, the size of the implants **10', 100**, etc., nodules **41** and hinges **43** can be varied to suit the needs of the individual patients. The implants of the present invention are not limited to any particular dimensions.

(45) To ensure that the inner surfaces of the top and bottom members **20, 30** of the implant **100** of the third preferred embodiment remain separatable once inserted into the interior volume **604**, preferably only a portion of the top and bottom members **20, 30** are placed into contact with one another during a majority of the demineralization. The remaining inner surfaces of the top and bottom members **20, 30** are separated from one another during demineralization to limit the

possibility that they do not become welded, secured or fixed together. For example, a wedge, masking or a barrier may be inserted between the inner surfaces of the top and bottom members **20**, **30** and, in particular, between the alternating projections **40** and recesses **45** formed in the top and bottom members **20**, **30**, to create a space between the top and bottom members **20**, **30** to limit the possibility that they do not become welded, secured or fixed together during demineralization.

(46) Referring to FIGS. 5A-5C, a flexible chain implant **200** in accordance with a fourth preferred embodiment includes a cap **240** to couple the top and bottom members **20**, **30** together. The top and bottom members **20**, **30** of the fourth preferred embodiment include a projection **256** comprised of a top projection **256a** and a bottom projection **256b** at the coupled portion **50** thereof when the top and bottom members **20**, **30** are nested together. That is, as shown, the top member **20** includes the top projection **256a** and the bottom member **30** includes the bottom projection **256b** such that when the top and bottom members **20**, **30** are nested together, the projection **256** is formed. The cap **240** preferably includes a recess **242** sized and configured to receive the projection **256** at the coupled portion **50** of the top and bottom members **20**, **30**. Moreover, as will be generally appreciated by one of ordinary skill in the art, the cap **240** may be configured to couple the top and bottom members **20**, **30** at another location.

(47) In use, the cap **240** is constructed and arranged to retain the coupled portion **50** of the top and bottom members **20**, **30** together so that the cap **240** fixedly secures the top member **20** to the bottom member **30** at the coupled portion **50** of the implant **200** even after the implant **200** has been inserted into the interior volume **604** of the targeted vertebral body **602**. However, the coupled portion **50** is not limited to fixedly securing the top and bottom members **20**, **30** together in the implanted configuration and may separate once the implant **200** is inserted into the interior volume **604**. By way of non-limiting example, the recess **242** formed in the cap **240** and the projection **256** formed at the coupled portion **50** of the nested top and bottom members **20**, **30** may be in the form a dovetailed joint. Alternatively, as will be generally appreciated by one of ordinary skill in the art, the cap **240** may include the projection **256** and the implant **200** may be formed with the recess **242**. When the cap **240**, top member **20** and bottom member **30** are constructed of the preferred bone material, the coupled portion **50** may be demineralized to secure or fix the cap **240** and projection **256** together at the coupled portion **50**.

(48) Referring to FIGS. 6A and 6B, the top and bottom members **20**, **30** of a flexible chain implant **300** in accordance with a fifth preferred embodiment includes a top end piece **322** and a bottom end piece **332**, respectively, having top and bottom apertures **324**, **334**, respectively. As shown, the top and bottom end pieces **322**, **332** are preferably located adjacent to the coupled portion **50** of the implant **300** but need not be located at the extreme end of the top and bottom members **20**, **30** in the insertion position. The top and bottom apertures **324**, **334** formed in the top and bottom end pieces **322**, **332** are sized and configured to receive a pin **360**. During assembly, the top end piece **322** and the bottom end piece **332** are preferably aligned such that the top aperture **324** and the bottom aperture **334** are aligned so that the top and bottom apertures **324**, **334** receive the pin **360** to secure and retain the top and bottom members **20**, **30** together. In use, the top and bottom end pieces **322**, **332** are constructed and arranged to retain the coupled portion **50** of the top and bottom members **20**, **30** together so that the pin **360** and top and bottom end pieces **322**, **332** fixedly secure the top member **20** to the bottom member **30** at the coupled portion **50** of the implant **300** even after the implant **300** has been inserted into the interior volume of the targeted vertebral body **602**. However, the coupled portion **50** is not limited to fixedly securing the top and bottom members **20**, **30** together in the implanted configuration and may separate once the implant **300** is inserted into the interior volume **604**.

(49) One of the top and bottom members **20**, **30** (shown as the top member **20**) may also include a stopper projection **326**. As shown, the stopper projection **326** is preferably formed in one or as part of one of the projections **40** formed on the implant **300**. In use, the stopper projection **326** acts as a physical barrier to prevent inadvertent removal of the pin **360**, particularly in the insertion position.

Moreover, the stopper projection **326** may be sized and configured to surround the head **362** of the pin **360** to limit exposure of the pin **360**. For example, as shown, the stopper projection **326** may include a recess **328** for receiving the head **362** of the pin **360**.

(50) Referring to FIGS. 7A and 7B, a flexible chain implant **400** in accordance with a sixth preferred embodiment includes a longitudinal extending projection **422** formed in the top member **20** adjacent to the coupled portion **50** while the bottom member **30** includes an end piece **432** having a corresponding longitudinal channel **434** for receiving the longitudinal extending projection **422**. The longitudinal extending projection **422** and the end piece **432** are preferably located adjacent to the coupled portion **50** of the implant **400** but need not be located at the extreme end of the top and bottom members **20**, **30** in the insertion position. Preferably, the longitudinal extending projection **422** is constructed and arranged for snap-fitting into the corresponding longitudinal channel **434** formed in the end piece **432** of the bottom member **30**. Once inserted, the longitudinal channel **434** is preferably constructed and arranged to retain the longitudinal extending projection **422** and hence limit separation of the top and bottom members **20**, **30** at the coupled portion **50**.

(51) In use, the longitudinal projection **422** and channel **434** are constructed and arranged to retain the coupled portion **50** of the top and bottom members **20**, **30** together so that the longitudinal projection **422** and channel **434** fixedly secure the top member **20** to the bottom member **30** at the coupled portion **50** of the implant **400** even after the implant **400** has been inserted into the interior volume **604** of the targeted vertebral body **602**. However, the coupled portion **50** is not limited to fixedly securing the top and bottom members **20**, **30** together in the implanted configuration and may separate once the implant **400** is inserted into the interior volume **604**. Alternatively, as will be generally appreciated by one of ordinary skill in the art, the bottom member **30** may include the longitudinal projection **422** and the top member **20** may include the end piece **432** and longitudinal channel **434**. When the top member **20** and bottom member **30** including the longitudinal projection **422** and channel **434** are constructed of the preferred bone material, the coupled portion **50** may be demineralized to further secure or fix the projection **422** and channel **434** together at the coupled portion **50**.

(52) Referring to FIGS. 8A and 8B, a flexible chain implant **500** in accordance with a seventh preferred embodiment includes a longitudinal extending projection **522** formed in the top member **20** adjacent to the coupled portion **50** of the implant **500** while the bottom member **30** includes an end piece **532** having a corresponding longitudinal channel **534** for receiving the longitudinal extending projection **522** formed on the top member **20**. The longitudinal extending projection **522** and the end piece **532** are preferably located adjacent to the coupled portion **50** of the implant **500** but need not be located at the extreme end of the top and bottom members **20**, **30** in the insertion position. Preferably, the longitudinal extending projection **522** is slideable into engagement with the corresponding longitudinal channel **534** formed in the end piece **532** on the bottom member **30**. Once inserted, the longitudinal channel **534** is preferably constructed and arranged to retain the longitudinal extending projection **522** and hence limit separation of the top and bottom members **20**, **30** at the coupled portion **50**.

(53) In use, the longitudinal extending projection **522** and channel **534** are constructed and arranged to retain the coupled portion **50** of the top and bottom members **20**, **30** together so that the longitudinal extending projection **522** and channel **534** fixedly secure the top member **20** to the bottom member **30** at the coupled portion **50** of the implant **500** even after the implant **500** has been inserted into the interior volume **604** of the targeted vertebral body **602**. However, the coupled portion **50** is not limited to fixedly securing the top and bottom members **20**, **30** together in the implanted configuration and may separate once the implant **500** is inserted into the interior volume **604**. Alternatively, as will be generally appreciated by one of ordinary skill in the art, the bottom member **30** may include the longitudinal extending projection **522** and the top member **20** may include the end piece **532** and the longitudinal channel **534**. When the top member **20** and bottom

member **30** including the longitudinal extending projection **522** and channel **534** are constructed of the preferred bone material, the coupled portion **50** may be demineralized to further secure or fix the projection **522** and channel **534** together at the coupled portion **50**.

(54) As previously stated, the flexible chain implants **10, 10', 100, 200, 300, 400, 500** of the preferred embodiments are constructed of bone. Alternatively, as will be described in greater detail below, the preferred implants **10, 10', 100, 200, 300, 400, 500** may be constructed of any biocompatible material now or hereafter known having the desired characteristics including, but not limited to, synthetic material, a biocompatible polymer, metal, ceramic, composite or any combination thereof. The preferred implants **10, 10', 100, 200, 300, 400, 500** may be absorbable or resorbable by the body and for specific applications may have osteoinductive properties or be made at least partly from osteoinductive materials. Accordingly, the preferred flexible chain implants **10, 10', 100, 200, 300, 400, 500** may be constructed of nearly any material or combination of materials that is able to take on the general shape and size and withstand or adapt to the normal operating conditions of the implants **10, 10', 100, 200, 300, 400, 500**.

(55) An exemplary method for fabricating a preferred flexible chain implant **10, 10', 100, 200, 300, 400, 500** from bone will now be described. In this example, a piece of stock allograft femoral bone, potentially the bone **13** shown in FIGS. **3A** and **3B**, is used as a base material, preferably, cortical allograft bone. Other bones may be used including, but not limited to, radius, humerus, tibia, femur, fibula, ulna, ribs, pelvis, vertebrae, etc.

(56) The selected bone **13** is preferably roughly machined to the desired general shape of the implant and/or components of the implant from the stock allograft bone **13**. For example, conventional milling, sawing, grinding and/or other fabrication techniques may be used. After machining the desired general shape, the implant and/or components of the implant are removed and/or separated from the raw material of bone **13** and excess material is removed. Additional machining is performed as necessary to remove any additional excess material and to form the desired shape and configuration of the implant and/or components of the implant. The shaped implant may be coupled and/or demineralized as necessary. The component portions of the implant are mounted together (see FIGS. **4A-8B**) and the preferred monolithic implants **10, 10'** or preferred, assembled multi-piece implants **100, 200, 300, 400, 500** are subjected to demineralization. The implants **10, 10', 100, 200, 300, 400, 500** are preferably demineralized until the hinges **43** are flexible enough to permit the implants **10, 10', 100, 200, 300, 400, 500** to move to and between the insertion position and the implanted configuration, such that the nodules **41** maintain general rigidity, at least at their core, and until the coupled portion **50** secure the top and bottom members **20, 30** of the multi-piece implants **100, 200, 300, 400, 500**. The amount of time and/or the concentration or composition of the demineralizing solution may be varied to provide the desired amount of flexibility, elasticity, or to weld, secure or fix a portion of the top and bottom members **20, 30** together. Demineralization may be applied to specific portions of the preferred implants **10, 10', 100, 200, 300, 400, 500**, for example, masking or shielding the portions that do not or should not be treated. For example, by masking portions intended to remain non-flexible, the remaining portions of the implants **10, 10', 100, 200, 300, 400, 500** may be partially or entirely demineralized, and the non-flexible portions may retain their original mineralized state prior to the masking.

(57) Various other configurations and methods for manufacturing monolithic, multi-piece or other coupled implants may be used. The choice of methods may depend, at least in part, on the material or materials to be used in the particular implant. For example, if the implant is made of a biocompatible polymeric material, the implant may be manufactured by using conventional manufacturing methods such as but not limited to milling and turning. Alternatively, if the implant is made out of a biocompatible polymeric material, the entire implant may be injection molded. If the implant is constructed of a metallic material, it may be manufactured using conventional manufacturing methods such as but not limited to milling and turning. However, the flexible components may undergo secondary processes such as annealing. The secondary process may be

limited to the flexible portions of the implant, for example by masking or shielding the non-flexible portions.

(58) Referring to FIGS. **10A** and **10B**, a flexible chain implant **1000'**, in accordance with eighth and ninth preferred embodiments, includes a plurality of bodies **1010** and a plurality of linking portions **1020** wherein each of the plurality of bodies **1010** includes a plurality of facets **1030**. The plurality of bodies **1010** includes a plurality of facets **1030** having or forming a hexagon shape in the eighth preferred embodiment and forming a pentagon shape in the ninth preferred embodiment.

Alternatively, the plurality of bodies **1010** may include a plurality of facets **1030** having or forming a polyhedral shape. It is to be understood that the shape of the plurality of bodies **1010** may be varied without deviating from the scope of the invention so long as the plurality of bodies **1010** are formed with a plurality of facets **1030** defining substantially flat surfaces.

(59) Forming each of the plurality of bodies **1010** with a plurality of facets **1030** facilitates stacking of the plurality of bodies **1010** in the implanted configuration. That is, a facet **1030** formed on one body **1010** may contact and come into facing engagement with a facet **1030** formed on another, adjacent body **1010** and reduce the amount of space present between two bodies **1010** as compared to other shapes, such as circular bodies, which tend to result in point contacts between the facing bodies in the implanted configuration. Moreover, because the implant **1000'** may bend, wrap, coil, etc., when inserted into the interior volume **604** of the targeted vertebral body **602**, non-adjacent bodies **1010** may also contact one another. Therefore, the reduction in gaps between the contacting bodies **1010** provided by forming the plurality of bodies **1010** with a plurality of facets **1030** may provide an increased density of bodies **1010** within the interior volume **604** of the targeted vertebral body **602**.

(60) Furthermore, forming the plurality of bodies **1010** with a plurality of facets **1030** may facilitate inserting the implant **1000'** into the interior volume **604** of the vertebral body **602**. That is, as previously described above in connection with the implants **10**, **10'**, **100**, **200**, **300**, **400**, **500** of the first through seventh preferred embodiments, generally, during insertion of the implant **1000'** into the interior volume **604** of the targeted vertebral body **602**, the implant **1000'** may be subjected to relatively large compressive stresses as the implant **1000'** is forced into the interior volume **604**. The plurality of bodies **1010** with a plurality of facets **1030** contact one another when the implant **1000'** is pushed into the interior volume **604** of the targeted vertebral body **602** via, for example, a cannula **606**. By providing the plurality of facets **1030** which come into facing engagement during insertion, forces on the facets **1030** are distributed over the facet **1030** surfaces. Moreover, by increasing the surface area in which bodies **1010** contact each other, the force at which the implant **1000'** may be inserted is increased.

(61) In the eighth and ninth preferred embodiments, the plurality of bodies **1010** of the implant **1000'** defines a longitudinal axis **1001** (e.g., a body axis) when the implant **1000'** is oriented in an extended position (FIGS. **10A** and **10B**). The plurality of linking portions **1020** preferably define a longitudinal axis **1022** (e.g., a link axis) in the extended position. The link axis **1022** is preferably off-set or off-centered from the body axis **1001** in the extended position (as shown in FIGS. **10A** and **10B**). Specifically, in the eighth and ninth preferred embodiments of at least FIGS. **10A** and **10B**, the link axis **1022** is preferably located a distance **A** from the body axis **1001**. By providing off-centered linking portions **1020**, the implant **1000'** may be partially restricted in its ability to bend in at least one plane, such as bending of the ends of the linking portions **1020** toward the opposite sides of the facets **1030**, because the relatively rigid facets **1030** bump into each other and restrict such bending. Restricting the ability of the implants **1000'** to bend in one plane enables the user to better control which way the implant **1000'** starts to bend and/or coil within the interior volume **604** of the targeted vertebral body **602**. However, as will be generally appreciated by one of ordinary skill in the art, the link axis **1022** may be coaxial with the body axis **1001**. In addition, the linking portions **1020** may permit twisting and misalignment of the facets **1030** relative to the body axis **1001**, thereby generally permitting bending and/or twisting of the individual facets **1030**

relative to each other in nearly any direction.

(62) Referring to FIGS. **11A-11C** in a tenth preferred embodiment, a flexible chain implant **1100** includes the plurality of bodies **1110** having tapered or curved surfaces **1112** such that the plurality of tapered or curved bodies **1110** have a narrow width **1114** and a wider width **1116**. Preferably, the plurality of linking portions **1120** interconnect the plurality of tapered bodies **1110** proximate the narrow width **1114**. The tapered or curved surfaces **1112** further include an abutment surface **1118** such that adjacent tapered or curved bodies **1110** may contact one another proximate the wider width **1116**. The plurality of abutment surfaces **1118** define a longitudinal axis **1119** (e.g., an abutment axis) in an extended position (FIGS. **11A** and **11B**), wherein the abutment axis **1119** is off-set or off-centered from a body axis **1101** in the extended position (as shown in FIG. **11A**). Specifically, in the tenth preferred embodiment, the abutment axis **1119** is preferably located a distance B from the body axis **1101** and off-set from a link axis **1122** in the extended position, as will be described in greater detail below. Preferably, the abutment axis **1119** is located on the opposite side of the longitudinal axis **1101** of the implant **1100** as compared to the link axis **1122**.

(63) During insertion of the implant **1100** into the interior volume **604** of the targeted vertebral body **602** via, for example, a cannula **606**, the implant **1100** may be subjected to relatively large compressive stresses as the implant **1100** is pushed into the interior volume **604** of the targeted vertebral body **602**. As a result of the plurality of bodies **1110** incorporating a tapered or curved surface **1112** which defines an abutment surface **1118**, adjacent bodies **1110** may contact one another, in effect, forming a solid implant that is capable of withstanding the compressive forces. Moreover, the cannula **606** is preferably sized and configured to snugly receive the implant **1100** therein so that the implant **1100** is braced by the inner surface of the cannula **606**. The sizing and arrangement of the plurality of bodies **1110** and the linking portions **1120** directs the majority of compressive forces, during advancement of the implant **1100** through the cannula **606** and into the interior volume **604** of the targeted vertebral body **602**, through the abutment surfaces **1118**. Providing a plurality of abutment surfaces **1118** may also enhance tactical feedback as the implant **1100** is pushed through the cannula **606**, as the implant **1100** generally feels like a solid implant to the technician conducting the procedure.

(64) The plurality of linking portions **1120**, which are thinner than the plurality of bodies **1110**, may be located off-centered, e.g., a longitudinal axis **1122** of the linking portions **1120** is preferably located a distance C from the longitudinal axis **1101** of the implant **1100** in the extended position. By providing off-centered linking portions **1120**, the implant **1100** may be restricted in its ability to bend in at least one plane. Restricting the implants **1100** ability to bend in one plane enables the user to better control which way the implant **1100** starts to bend and/or coil within the interior volume **604** of the targeted vertebral body **602**. However, as will be generally appreciated by one of ordinary skill in the art, the longitudinal axis **1122** of the linking portions **1120** may be coaxial with the longitudinal axis **1101** of the implant **1100**.

(65) Referring to FIG. **11C**, in the tenth preferred embodiment, the shape and configuration of the implant **1100** is configured for harvesting from a single, cylindrical section of bone, generally to improve the yield of the specific bone. Specifically, if a particular bone is not relatively long or does not include a relatively long portion to accommodate construction of the relatively long, flexible chain implants **10**, **10'**, **100**, **200**, **300**, **400**, **500** and/or components of the first through seventh preferred embodiments, the implant **1100** of the tenth preferred embodiment may be rough machined in the generally cylindrical configuration, shown in FIG. **11C**. The implant **1100** is then preferably subjected to demineralization such that the linking portions become flexible and the bodies **1110** maintain some rigidity. Following demineralization, the implant **1100** may be oriented in the expanded position and/or inserted in the cannula **606** in the insertion position. In addition, the implant **1100** may be urged through the cannula **606** into the interior volume **604** and fold, wind and generally, randomly position itself in the implanted configuration.

(66) Referring to FIGS. **1A-13B**, a minimally invasive method of augmenting a damaged vertebral

body **602**, e.g., following a VCF, may include inserting one or more of the preferred flexible chain implants **10, 10', 100, 200, 300, 400, 500, 1000', 1100** into the interior volume **604** of the targeted vertebral body **602** between the endplates of the vertebral body **602**. One or more flexible chain implants **10, 10', 100, 200, 300, 400, 500, 1000', 1100** may be inserted as a preventive measure to augment the vertebral body **602** before compression or a compression fracture. For example, generally, a passageway **610** may be formed in the outer cortical shell of the targeted vertebral body **602** by a trocar, drill or other instrument. By way of non-limiting example, the passageway **610** may be formed into the interior of the vertebral body **604** through a posterior portion of the vertebral body, such as, for example, through a pedicle **608a, 608b**. The cannula **606** may then be introduced into the interior volume **604** of the targeted vertebral body **602** through the passageway **610**. The implant **10, 10', 100, 200, 300, 400, 500, 1000', 1100** is inserted into the interior volume **604** of the targeted vertebral body **602** through the cannula **606** and into interior volume **604**. The implant **10, 10', 100, 200, 300, 400, 500, 1000', 1100** may be inserted into the interior volume **604** by, for example, a push rod. Utilization of the push rod is particularly useful for inserting the preferred implants **10, 10', 100, 200, 300, 400, 500, 1000', 1100**, as the implants **10, 10', 100, 200, 300, 400, 500, 1000', 1100** are designed to withstand the relatively large compressive forces that may be experienced during insertion.

(67) Alternatively, as shown in FIGS. **12A-12F**, an exemplary embodiment of a cannula **1200** may incorporate an advancement mechanism **1210**. As best shown in FIGS. **12A-12E**, the advancement mechanism **1210** may be in the form of a cork-screw. The cork screw is preferably adapted for inserting the flexible chain implants **1000, 1000'** and **1100** of the prior art and the eighth and ninth preferred embodiments. That is, the advancement mechanism **1210** may include a section of spiraling geometry **1212**. The spiraling geometry **1212** includes an inner diameter, an outer diameter, a pitch, and a cross-sectional thickness and shape. The spiraling geometry **1212** resembles a spring (with a cross section that is circular, square, rectangular, etc.), an auger, a worm gear, or a screw. Using the spiraling geometry **1212**, the implants **1000, 1000', 1100** could be fed or weaved into the individual coils or spirals of the spiraling geometry **1212** of the advancement mechanism **1210**. As shown in FIG. **12C**, the proximal end **1214** of the spiraling geometry **1212** may initially start at a larger diameter which may taper down to the functional diameter to facilitate loading of the implants **1000, 1000', 1100** into the advancement mechanism **1210**. Loading may be done through an opening in the side of the cannula **1200** or the implants **1000, 1000', 1100** may be wound around the end of the advancement mechanism's spirals.

(68) The spiraling geometry **1212** may be off-axis to the longitudinal axis of the implants **1000, 1000', 1100**. Alternatively, as best shown in FIGS. **12D** and **12E**, the implants **1000, 1000', 1100** may have a longitudinal axis that is substantially coincident with the longitudinal axis of the spiraling geometry **1212** within the cannula **1200**.

(69) As best shown in FIG. **12A**, the cannula **1200** may also include a handle **1230**, which may be detachable to facilitate surgeon placement via hand or mallet impaction, and an optional extended section **1220** between the handle **1230** and the advancement mechanism **1210**. The cannula **1200** may also include a locking mechanism to help prevent separation of the advancement mechanism **1210** from the cannula **1200**. The locking mechanism preferably is capable of being disengaged as part of normal operation of the instrument and/or to facilitate sterilization. The cannula **1200** is also preferably sized and configured to permit the implants **1000, 1000', 1100** to travel down through the cannula **1200**. Accordingly, the cannula **1200** is sized and configured to prevent the implants **1000, 1000', 1100** from rotating in place with the advancement mechanism **1210**. For example, a mechanism for preventing the implants **1000, 1000', 1100** from rotating in place may be formed within the sidewall of the cannula **1200**. For example, the inner sidewall of the cannula **1200** may include a longitudinal slot or track that is sized and configured to mate with at least a portion of the implants **1000, 1000', 1100**. Alternatively, the cannula **1200** may have a non-circular-shaped lumen. For example, the cannula **1200** may have an oval shaped inner diameter which tightly matches the

geometry of the implants **1000**, **1000'**, **1100**. This eliminates the necessary space needed to allow the implants **1000**, **1000'**, **1100** to rotate in place.

(70) As best shown in FIG. **12F**, the cannula **1200** may also include an adjustable skin stop attachment **1240** to help maintain the cannula **1200** at the desired depth within the vertebral body **602**. A small venting shaft may also be used along the side of the cannula **1200** in order to vent any air that may get forced into the operating site. This could be especially important if the same cannula **1200** is used to inject liquid bone void filler. If the cannula **1200** is to be used to inject liquid bone void filler, a lure lock feature may be preferred on the cannula **1200** to connect it to the filler's injection device. In the event that a portal is included in the side of the cannula **1200** as an access point to couple the implants **1000**, **1000'**, **1100** within the spiraling geometry **1212**, a cap or door may be provided to snap on, hinge, or slide over the loading portal to help retain the coupling.

(71) The implant is preferably inserted into the vertebral body **602** in an insertion position and is oriented in an implanted configuration in the vertebral body **602** so that the implant augments the interior volume **604** of the targeted vertebral body **602** so that large pockets or voids are limited within the interior volume **604**, while small voids amenable to boney in-growth and providing a vascular pathway are provided. The implant preferably separates or unwraps from the insertion position and, generally, randomly coils and wraps or twists around itself in the implanted configuration.

(72) Insertion of the implant into the interior volume **604** of the targeted vertebral body **602** may compact the cancellous and/or osteoporotic bone inside the vertebral body **602** and/or restore or support the vertebral body **602**. Alternatively, after the passageway **610** is formed in the vertebral body **602**, instruments such as, for example, curettes or a balloon catheter may be used to compress and compact the bone inside the vertebral body **602** to create the interior volume **604**. The instruments may then be removed. Alternatively, the balloon portion of the catheter may remain within the vertebral body **602** or may form a container for the implant. The interior volume **604** in the vertebral body **602** also may be formed by removing bone material as opposed to compacting the bone. For example, a reamer or other apparatus could be used to remove bone material from the inside of the vertebral body **602**. Further, the interior volume **604** may be formed by urging the implant into the body **602** to restore vertebral body height following a VCF.

(73) Whether a cavity is first formed in the targeted vertebral body **602** or the implant is inserted without first creating a cavity, as one or more implants or portions thereof are inserted into the interior volume **604** of the targeted vertebral body **602**, the implant augments the interior volume **604** and provides structural support to stabilize the vertebral body **602**. In a vertebral body **602** that has collapsed, as the implant augments the interior volume **604**, the implant may push against the inner sides of the endplates, thereby tending to restore the vertebral body **602** from a collapsed height to its original or desired treated height and provide structural support to stabilize the vertebral body **602**. Instead of using the insertion of the implant to restore the height of the vertebral body **602**, an instrument may be inserted through the passageway to restore the height of the vertebral endplates. For example, a balloon catheter may be inserted to restore the vertebral endplates, or an elongated instrument that contacts the inside of the endplates and pushes on them may be utilized. These techniques may also be utilized in combination to restore height of the vertebral body **602**.

(74) The flexibility of the implant allows bending of the implant within the interior volume **604**, e.g., in a non-uniform or tortuous configuration, to aid in ensuring a thorough integration of the implant within the interior volume **604** of the targeted vertebral body **602**, or potentially in a uniform pattern. The flexibility of the implant, as well as the configuration of the implant, may also permit bending of the implant to augment the interior volume **604** of the targeted vertebral body **602**. The separation of the top and bottom members **20**, **30** may also allow the implants to collapse and possibly become entangled so that it becomes larger than its insertion hole so that it cannot be easily ejected.

(75) PMMA or another bone cement or filler (for example bone chips) may be inserted sequentially or simultaneously into the interior volume **604** of the targeted vertebral body **602** along with the implant to further enhance fixation or repair of the targeted vertebral body **602**. Alternatively and/or in addition, a plug (not shown) of bone cement may be inserted into the passageway **610** that was initially formed to insert the implant. The plug may cover the insertion passageway **610** to prevent the implant from being removed or ejected. In other embodiments, some or all of the implant may be removed after repositioning the targeted vertebral body **602**, and PMMA or another bone cement or filler may be injected into a void created by the implant. Alternatively a bone growth promoting filler may be inserted into the interior volume **604** and a plug of bone cement may be utilized to hold the implant and filler material in the vertebral body **602**.

(76) In some embodiments, the implant may be coated with an adhesive, such that the implant may be inserted into the targeted vertebral body **602** in a flexible state and may become tangled and/or convoluted during or after insertion. After insertion, the implant may become attached together by the adhesive so that the implant becomes a mass that may be locked into the interior volume **604** of the targeted vertebral body **602**, or otherwise secured such that the implant may not be easily removed through the passageway **610**.

(77) In other embodiments, the implant may be coated with an adhesive and the implant may be inserted, with or without becoming tangled or convoluted, into the interior volume **604** of the targeted vertebral body **602**. During or after insertion, some or all of the implant may be exposed to an energy source (e.g., an ultraviolet light, ultrasonic radiation, radio waves, heat, electric field, magnetic field), for example to activate the adhesive, such that the exposed portion of implant becomes joined to form a mass, or becomes rigid, or both, thereby further augmenting the vertebral body **602** and/or preventing removal or ejection of the implant through the insertion opening.

(78) Although the various embodiments of the flexible chain implants and methods described herein thus far have been described in the context of repositioning and augmenting a vertebral body **602**, for example in the context of VCF and deformations in spinal curvature, various other uses and methods are envisioned. For example, in some embodiments, the implant may be used to augment a vertebral body **602** where a compression or a compression fracture has not yet occurred and thus may be preventative in nature. Also, in some embodiments the implant may be used between two vertebrae. For example, the implant may be inserted in the annulus of a spinal disc, or the disc may be removed and the implant inserted between adjacent vertebrae to promote fusion of adjacent vertebrae. The implant in some embodiments may be insertable in an additional implant, such as a cage implanted between adjacent vertebrae. The implant may also be used to reposition and/or augment other damaged bone regions such as a fractured or weakened proximal femur.

(79) In some embodiments, the various embodiments of the flexible chain implants and methods described herein may be used in conjunction with other apparatus and methods to restore lordosis and augment the vertebral body **602**. For example, one or more implants may be used in conjunction with known procedures that may be used to begin repositioning of a vertebral body **602** and/or to create a space within the vertebral body **602** for the implant. In other embodiments, one or more implants may be used in conjunction with other tools or external fixation apparatus for helping to manipulate or fix the vertebral body **602** or other bones in a desired position.

(80) In another embodiment, a kit including various combinations of assemblies and components may include, for example, a cannula **1220**, **606** or other introducer and one or more implants. The one or more implants may be provided in different sizes, e.g., different lengths and/or diameters. In other embodiments, the kit may include one or more of the following: an introducer, one or more implants, a syringe or other apparatus for injecting a cement or other filler into a vertebral body or other space, one or more balloon catheters, curettes, and other instruments and may additionally include anchoring elements, tensioning members, fixation members, or any combination thereof. One skilled in the art will appreciate that various other combinations of devices, components and assemblies may be made and are intended to fall within the scope of the present invention.

(81) As described herein, the implants are preferably made from bone, more preferably allograft material, although use of xenograft and autograft is also envisioned. Furthermore, the implants described herein may be manufactured from materials with varying levels of porosity, such as by combining bone sections from different bones or different types of tissues and/or materials having varying levels of porosity.

(82) Moreover, the implants, described herein, may be manufactured from bone materials having varying mineral content. For example, cancellous or cortical bone may be provided in natural, partially demineralized, or demineralized states. Variation in the mechanical properties of the bone sections used may be obtained through various amounts of demineralization. Advantageously, use of a demineralizing agent on bone, e.g., cortical or cancellous bone, transforms the properties of the bone from a stiff structure to a relatively pliable structure. Optionally, the flexibility or pliability of demineralized bone may be enhanced when the bone is hydrated. Any desired portions of the bone components may be demineralized or partially demineralized in order to achieve a desired amount of malleability, elasticity, pliability or flexibility, generally referred to herein as “flexibility.” The amount of flexibility can be varied by varying in part the amount of demineralization.

(83) In some embodiments, flexibility of demineralized or partially demineralized regions may be further enhanced by varying the moisture content of the implant, or portions thereof. Bone components initially may be provided with moisture content as follows: (a) bone in the natural state fresh out of the donor without freezing, (b) bone in the frozen state, typically at negative forty degrees Celsius (-40°C), with moisture content intact, (c) bone with moisture removed such as freeze-dried bone, and (d) bone in the hydrated state, such as when submersed in water. Using the expansion and contraction properties that can be obtained during heating and cooling of the bone material, and the concomitant resorption of moisture along with swelling for some bone material, permits alternate approaches to achieving a desired flexibility of the implant within a bone or other region.

(84) The implants may be formed entirely from cortical bone, entirely from cancellous bone, or from a combination of cortical and cancellous bone. While the implants may be created entirely from all bone material, it is also anticipated that one or more components or materials may be formed of non-bone material, including synthetics or other materials. Thus, while the implants, described herein, are typically described as being formed primarily from bone, the implants alternatively may be manufactured, in whole or in part, from any biocompatible material known in the art including, but not limited to, metals (such as, for example, stainless steel or titanium), alloys, hydroxyapatite, resorbable material, ceramics, polymers, composites, and encapsulated fluids, gels, etc.

(85) One particular preferred alternate embodiment for manufacturing the implants described herein is a synthetic material. If the base synthetic material is naturally radiolucent then a radiopaque agent such as BaSO_4 could be blended into the base material to make it visible on, for example, a C-arm image. The thin sections need only be thick enough so as to act as flexible “living hinges”. Living hinge technology may be applied to the implants using a huge variation of plastic resins, such as polyethylene (PE), polypropylene (PP), and polycaprolactone (PCL). Plastics may offer the ability to be porous to promote bony in-growth or on-growth as well the ability to impregnate the base resin with pockets of chemicals, vitamins, medication, etc. that can release over time. Some resins are bioresorbable, such as, for example, PCL but others are not, such as, for example, PE, and it is possible that both characteristics could be useful in a bone void filler application depending on the indications. There are also some super-elastic metals such as Nitinol that may offer acceptable flexibility if the thin sections are designed small enough to be flexible and the thick sections are designed thick enough to add structural support to the tangled implant mass. Further, it is likely that a wide range of geometries and features can be included by using a synthetic material.

(86) Additional applications of the above described implants can be realized in burst fractures,

wherein extrapedicular or transpedicular placement of the implants can add stability during healing; prophylactic applications, wherein extrapedicular or transpedicular placement of the implants can be used within an intact (unfractured) vertebral body adjacent to a long-segment fusion or adjacent to an alternative bone stabilization construct, to provide additional stability as well as a more natural stiffness transition to the untreated levels when compared to the use of PMMA cement; interbody fusion extender, wherein placement of the implants within intervertebral spacers such as, for example, PLIF and TPLIF spacers to assist in fusion; and percutaneous placement of the implants within the lateral gutters to assist in fusing the posterolateral elements such as the adjacent transverse processes; and back filler material for iliac-crest autograft sites to help speed along the healing at this location and to reduce donor site morbidity.

(87) While the foregoing description and drawings represent the preferred embodiments of the present invention, it will be understood that various additions, modifications and substitutions may be made therein without departing from the spirit and scope of the present invention as defined in the accompanying claims. In particular, it will be clear to those skilled in the art that the present invention may be embodied in other specific forms, structures, arrangements, proportions, and with other elements, materials, and components, without departing from the spirit or essential characteristics thereof. One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, materials, and components and otherwise, used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. In addition, features described herein may be used singularly or in combination with other features. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being indicated by the appended claims, and not limited to the foregoing description.

(88) It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that this invention is not limited to the particular embodiments disclosed, but it is intended to cover modifications within the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A method of implanting a flexible chain implant, the method, comprising: loading a flexible chain implant into a cannula of an introducer, wherein the loading step comprises weaving a plurality of bodies of the flexible chain implant into a plurality of coils of an advancement mechanism insertable within the cannula, the flexible chain implant having a plurality of links that interconnect the plurality of bodies, wherein the advancement mechanism is elongate along a central axis, and the plurality of coils define an inner diameter that is spaced from the central axis such that the plurality of coils are open between the central axis and the inner diameter, wherein the inner diameter diminishes in a distal direction from a proximal end of the advancement mechanism to a distal end thereof; inserting a distal end of the introducer into an interior volume within bone; and rotating the advancement mechanism within the cannula, thereby advancing the flexible chain implant through the cannula and outward from the distal end of the introducer and into the interior volume.
2. The method of claim 1, wherein the plurality of bodies defines a body axis that extends through centers of the bodies when the flexible chain implant is loaded within the cannula, the plurality of links defines a link axis that extends through centers of the flexible links when the flexible chain implant is loaded in the cannula, and the link axis is offset from the body axis within the cannula by a distance during the rotating step.
3. The method of claim 1, further comprising attaching a handle to a proximal end of the cannula

before the inserting step.

4. The method of claim 3, wherein the cannula includes an elongate portion extending from the proximal end of the cannula to the advancement mechanism.

5. The method of claim 1, further comprising, during the rotating step, engaging the flexible chain implant with a mechanism formed along an interior surface of a sidewall of the cannula in a manner preventing the flexible chain implant from rotating along with the advancement mechanism.

6. The method of claim 5, wherein the mechanism comprises a track extending along a longitudinal direction that is parallel with the central axis, and the engaging step comprises mating at least a portion of the plurality of bodies with the track.

7. The method of claim 6, wherein the track is a slot formed in the interior surface.

8. The method of claim 5, wherein the mechanism comprises a shape of the interior surface, wherein the shape matches a geometry of the plurality of bodies.

9. The method of claim 1, wherein the loading step comprises feeding the plurality of bodies into the cannula through a side opening of the cannula.

10. The method of claim 1, further comprising communicating bone filler material distally through the cannula and outward from the distal end of the introducer an into the interior volume within bone.

11. The method of claim 1, further comprising adjusting a position of a skin stop mechanism on an outer surface of the cannula, and subsequently abutting the skin stop mechanism against skin of the patient in a manner maintaining the distal end of the introducer at a desired depth within the bone.

12. The method of claim 1, wherein the plurality of links are flexible to enable adjacent or non-adjacent bodies of the plurality of bodies to contact one another within the interior volume within bone.

13. The method of claim 1, wherein: each of the plurality of bodies of the flexible chain implant defines a plurality of exposed facets, and the plurality of exposed facets are rigid for facilitating stacking the plurality of bodies within the interior volume within bone, and the plurality of facets of each body includes: a center facet that extends distally while advancing distally through the cannula; and a plurality of surrounding facets, each sharing a different common edge with the center facet.

14. The method of claim 13, wherein the center facet has at least five edges, and the plurality of surrounding facets comprise at least five surrounding facets each sharing a different one of the at least five edges with the center facet.

15. The method of claim 13, wherein the plurality of bodies defines a body axis that extends through centers of the bodies when the flexible chain implant is disposed within the cannula, the plurality of links defines a link axis that extends through centers of the flexible links when the flexible chain implant is disposed in the cannula, and within the cannula the link axis is offset from the body axis by a distance during the rotating step.
